Project Overview & Goals

The Python package `fleet_management` schedules a fleet of F trains over a horizon of T days. Every day, M missions must be flown. Each train flies at most one mission, spends the day in the depot, or stays idle. Each train has L components, and each component degrades stochastically. On a depot day a component may be repaired (imperfect repair) or replaced. The schedule is a mixed-integer program solved with Gurobi.

Goal. Minimise the cost of depot days, repairs and replacements, plus a penalty on the largest train damage (Section 4), under a reliability requirement: for every component and every day, $\Pr(D_{i\ell k} > \tau_{i\ell}) \le \varepsilon_{i\ell}$, with $D_{i\ell k}$ the degradation and $\tau_{i\ell}$ the failure threshold. The horizon has two phases. The transitory phase (H_1 days) starts from the current state of the fleet. The operating phase (H_2 days) is the schedule to repeat. The requirement holds on every day of the horizon and of every later repetition of the operating phase (Section 5.9).

Degradation models. Each cell (component ℓ of train i) has its own model, chosen by the input `model`:

- **Gaussian**: a Gaussian random walk; tracks the mean and the variance, and a repair reduces both (Section **??**).
- **Inverse Gaussian (IG)**: tracks the mean; reliability by a cap on the mean (Section **??**).
- **Wiener**: Brownian motion with drift; tracks the mean and the variance. The variance grows by σ^2 per mission day; a repair acts on the mean only, and only a replacement lowers the variance (Section **??**).
- **Gamma**: Gamma increments with mission-dependent rates; tracks the mean; reliability by a cap that is safe for every mix of rates (Section **??**).
- **Rainflow**: distribution-free. The increment law is measured by rainflow counting of load histories and enters only through summaries (mean, variance, support width, cumulant generating function). Reliability is certified by one of five tail bounds: Markov, Cantelli, Hoeffding, Bernstein or Chernoff (Section **??**).

Both the Gaussian and the Wiener model use Gaussian increments. The Gaussian model tracks a variance that a repair reduces. The Wiener model fits components whose uncertainty does not depend on the route and is not reduced by a repair. When in doubt, use the Gaussian model. The rainflow model makes no distributional assumption: its certificate holds for every increment law with the measured summaries, at the price of a looser bound than an exact distribution function.

Repair and replacement. Each cell has a repair law, ARD_∞ (the default) or ARD_1 (Section 5.3); every model accepts both. Each cell may be replaced (Section 5.5). The loop row forces at least one replacement in the operating phase in two cases. This is not an input check; a schedule without the replacement is infeasible.

- A Wiener cell whose train flies at least one mission in the operating phase: its variance falls only by replacement.
- An ARD_1 cell with $\rho_{i\ell} < 1$ that accumulates mean damage in the operating phase: a repair never removes the share $1 - \rho$ of the damage since the last intervention (Proposition 5.8).

Problem class. The problem is a mixed-integer linear program (MILP) when every reliability row is linear and the damage penalty is `inf_norm`. Otherwise it is a mixed-integer quadratically constrained program (MIQCP). It is nonconvex when a quadratic reliability row uses the `exact` form (Section 5.6); it is convex when the only quadratic row is the `quadratic` damage penalty (Section 5.8).

Scope of the package. The package solves one schedule (`solve`), sweeps the horizon lengths (Section **??**), starts a solve from a previous solution or a heuristic (Section **??**), plots a schedule

(Section **??**), and checks a result against its input (Section **??**). It reads inputs in YAML, JSON or HDF5 (Section **??**) and writes the outputs of Section **??**.

Authority. This specification is authoritative for the formulation and for the tools. The code follows it: every change is written here first.

Non-goals.

- Routing decisions are not optimised: the missions are inputs.
- Continuous-time formulations are out of scope: all dynamics are discrete-time, with a step of one day.

Index Sets and Notation

Symbol	Meaning	Defined in
Indices and horizon		
$i \in \{1, \dots, F\}$	train	
$\ell \in \{1, \dots, L\}$	component of a train	
(i, ℓ)	a *cell*: component ℓ of train i	
$j \in \{0, 1, \dots, M\}$	activity: $j = 0$ is a depot day, $j \geq 1$ a mission	
$k \in \{1, \dots, T\}$	day; index $k = 0$ holds the initial state	
$H = [H_1, H_2]$, $T = H_1 + H_2$	horizon input; a scalar H means $H_1 = H_2 = H$	**??**
days $1, \dots, H_1$	transitory phase	
days $H_1 + 1, \dots, T$	operating phase, repeated without end in the endless schedule	5.9
$p \in \{1, \dots, P\}$	segment of the `pwl` form, $P =$ `pwl_points`	5.6
Parameters, increments and states		
$\tau_{i\ell}$, $\varepsilon_{i\ell}$, $\rho_{i\ell}$	failure threshold, reliability level, repair efficiency of a cell; a model section may drop the subscript $i\ell$ after saying so	**??**
C_M; $C_{R,i\ell}$, $C_{\mathrm{rep},i\ell}$; C_D	costs: depot day; repair, replacement; damage penalty	4
$\mu^{\mathrm{inc}}_{i\ell jk}$, $v^{\mathrm{inc}}_{i\ell jk}$, $b^{\mathrm{inc}}_{i\ell jk}$, $K^{\mathrm{inc}}_{i\ell jk}$, $\beta_{i\ell jk}$	mean, variance, support width and cumulant generating function (CGF) of the increment of cell (i, ℓ) when train i flies mission j on day k, after the phase lookup; Gamma rate	**??**
$\delta\mu_{i\ell k} = \sum_{j=1}^{M} x_{ijk}\, \mu^{\mathrm{inc}}_{i\ell jk}$	day increment of the mean; δv, δR (from $(b^{\mathrm{inc}})^2$) and δK likewise; 0 on depot and idle days	
$\mu_{i\ell k}$, $v_{i\ell k}$, $R_{i\ell k}$, $K_{i\ell k}$	tracked mean, variance, Hoeffding budget, Chernoff quantity	3
$g^{\mu}_{i\ell k}$, $g^{v}_{i\ell k}$, $g^{R}_{i\ell k}$, $g^{K}_{i\ell k}$	anchors (law ARD_1)	5.3
$\mu_{0,i\ell}$, $v_{0,i\ell}$, $R_{0,i\ell}$, $K_{0,i\ell}$	initial values	**??**
$\mu^{\mathrm{new}}_{i\ell}$, $v^{\mathrm{new}}_{i\ell}$, $R^{\mathrm{new}}_{i\ell}$, $K^{\mathrm{new}}_{i\ell}$	replacement values	5.5
$b^{0}_{i\ell}$, $b^{\mathrm{new}}_{i\ell}$	support widths of the initial and replacement damage	**??**
ARD_∞, ARD_1; a; ψ	repair laws; repair factor; repair map	5.3
ξ_k, ζ_k, φ_k, Φ, $\mathcal{C}_{i\ell}$	full tracked state, reference trajectory, day map, map of the operating phase, certificate set	5.4, 5.9
Derived constants		
M_s	natural range of a tracked quantity s (big-M constant)	5.4
$\Phi^{-1}(1-\varepsilon)$	standard normal quantile	**??**
$V_{\max} = \tau^2/(\Phi^{-1}(1-\varepsilon))^2$	variance range, Gaussian and Wiener	**??**
$\eta_{i\ell}$; $Z \sim \mathrm{IG}(1, 1/\eta)$; q; $\bar{\mu} = \tau/\mathbb{E}[Z \mid Z > q]$	inverse Gaussian dispersion; unit law; its $(1-\varepsilon)$-quantile; mean cap	**??**
$\beta^{*}_{i\ell}$; $t = \beta^{*}\tau$; $A_1(t, \varepsilon)$; $Q(a, x)$	smallest Gamma rate among the terms of the cell with a positive mean; scaled threshold; shape cap; regularised upper incomplete gamma function	**??**
$\kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}$	Cantelli coefficient	**??**

Symbol	Meaning	Defined in
$\kappa = \ln(1/\varepsilon)$	exponent constant (Hoeffding, Bernstein)	??
$b_{i\ell}$, $c_{i\ell} = \kappa\, b_{i\ell}/3$	Bernstein support constant and margin	??
$s_{i\ell}$	Chernoff parameter (input `s_chernoff`)	??
V^{C}, V^{B}	variance ranges, rainflow Cantelli and Bernstein	??
$\mu_r = \texttt{tangent_ref} \cdot \tau$; c_2, c_1	tangent point; constants of a quadratic row	5.6
E	evaluation horizon (input `evaluation_horizon`)	4

Decision Variables

A variable, and the rows that define it, are built only where a row or an objective term with a non-zero coefficient uses them. Example: u and its rows are not built when $C_D = 0$; $z_{i\ell k}$ and its rows are not built at a cell with $C_{R,i\ell} = 0$. The day index runs over $k = 1, \ldots, T$; index $k = 0$ holds constants (initial values and starting anchors), not variables.

Variable	Shape, type	Built when	Meaning
x_{ijk}	$F \times (M{+}1) \times T$, binary	always	1 if train i flies mission j on day k; $j = 0$: depot day
$x^m_{i\ell k}$	$F \times L \times T$, binary	always	1 if component ℓ of train i is repaired on day k
$x^r_{i\ell k}$	$F \times L \times T$, binary	`allow_replacement = true`	1 if component ℓ of train i is replaced on day k
$n_{i\ell k}$	binary	cells with `encoding = "indicator"`	1 on a day without intervention, Eq. (15)
$\mu_{i\ell k}$	continuous, $[0, \tau_{i\ell}]$	every cell	tracked mean degradation
$v_{i\ell k}$	continuous, $[0, M_v]$	cells whose model or bound tracks a variance	tracked variance
$R_{i\ell k}$	continuous, $[0, M_R]$	rainflow cells with the Hoeffding bound	tracked Hoeffding budget
$K_{i\ell k}$	continuous, $[0, M_K]$	rainflow cells with the Chernoff bound	tracked Chernoff quantity
$g^{\mu}_{i\ell k}$, $g^{v}_{i\ell k}$, $g^{R}_{i\ell k}$, $g^{K}_{i\ell k}$	continuous, range of the quantity	ARD_1 cells, one per tracked quantity except the Wiener variance	anchors, Eq. (10)
$w_{i\ell kp}$	binary, $p = 1, \ldots, P$	cells with a quadratic row in the `pwl` form, $P \geq 2$	segment of the mean, Eq. (20)
$z_{i\ell k}$	continuous, ≥ 0	cells with $C_{R,i\ell} > 0$, on days whose cost enters the objective	degradation removed by a repair, Section 5.7
u	scalar, ≥ 0	$C_D > 0$	damage regularisation, Section 5.8

The ranges M_v, M_R, M_K are the natural ranges of Section 5.4; each model section states which quantities it tracks. The output reports the values of the tracked quantities; they are upper bounds on the reference values (Section 5.4).

Objective Function

The cost of day k is

$$c_k = C_M \sum_{i=1}^{F} x_{i0k} + \sum_{i=1}^{F} \sum_{\ell=1}^{L} C_{R,i\ell}\, z_{i\ell k} + \sum_{i=1}^{F} \sum_{\ell=1}^{L} C_{\mathrm{rep},i\ell}\, x^r_{i\ell k}, \tag{1}$$

where $z_{i\ell k}$ reads 0 where it is not built, and the last sum is absent when `allow_replacement = false`. The costs of the two phases are

$$J_{\text{init}} = \sum_{k=1}^{H_1} c_k, \qquad J_{\text{op}} = \sum_{k=H_1+1}^{T} c_k. \tag{2}$$

The input `objective_mode` (default `total`) selects the objective:

$$\min \begin{cases} J_{\text{init}} + J_{\text{op}} + C_D\, u & \texttt{total}, \\ J_{\text{op}}/H_2 + C_D\, u & \texttt{operating_average}, \\ J_{\text{init}} + \dfrac{E - H_1}{H_2}\, J_{\text{op}} + C_D\, u & \texttt{evaluation_total}, \end{cases} \tag{3}$$

where E is the input `evaluation_horizon`, required in the mode `evaluation_total`, with $E > H_1$. The term $C_D\, u$ is present only when $C_D > 0$ (Section 5.8).

Cost terms.

- $C_M \geq 0$, a scalar, is paid once per train per depot day ($x_{i0k} = 1$), whatever the number of components repaired or replaced on that day. It prices the visit to the depot, for example the train out of service and the set-up. An idle day (Section 5.1) costs nothing, so the number of depot days is a costed decision.
- $C_{R,i\ell} \geq 0$ is paid per unit of mean degradation removed by a repair (Section 5.7).
- $C_{\text{rep},i\ell} \geq 0$ is paid per replacement. The input is required; it has no default.
- $C_D \geq 0$ weights the damage penalty u (Section 5.8). The term is added once in every mode, because u is the maximum over the whole horizon.

The shapes of the cost inputs are given in Section **??**.

Meaning of the modes.

- `total`: the cost of the T modelled days.
- `operating_average`: the cost per day of the operating phase. It is the long-run cost per day of the endless schedule (Section 5.9), in which the cost of the transitory phase spreads over infinitely many days. It compares solves with different H_2 (Section **??**). The transitory costs do not enter the objective in this mode: the solver chooses the transitory schedule for feasibility only.
- `evaluation_total`: the projected cost over E days: the transitory phase once, then $(E - H_1)/H_2$ repetitions of the operating phase, a fraction of a repetition counted pro rata. It also prices the transitory phase, so it compares solves with different H_1.

Example: $H_1 = 5$, $H_2 = 10$, $E = 365$. Then `evaluation_total` weights the operating cost by 36, and `operating_average` by 1/10.

The objective value is computed from the tracked values. It is therefore an upper bound on the cost of the schedule computed from the reference values (Section 5.4).

Constraints

The constraints have three groups. The fleet rows (assignment and depot days) do not depend on the degradation models. The common rules of the dynamics (repair laws, encoding, replacement, linear forms of the reliability rows) hold for every model; the five model sections apply them and add each model's tracked quantities, day maps and reliability row. The repair cost, the damage regularisation and the loop constraint close the problem. All rows hold for every day $k = 1, \ldots, T$ unless a section says otherwise.

Assignment Constraints

Each train performs at most one activity per day:

$$\sum_{j=0}^{M} x_{ijk} \leq 1 \qquad \forall\, i, k. \tag{4}$$

Each mission is flown by exactly one train every day:

$$\sum_{i=1}^{F} x_{ijk} = 1 \qquad \forall\, j \in \{1, \ldots, M\},\ k. \tag{5}$$

Together these rows put exactly M trains on missions every day. The other $F - M$ trains are in the depot or idle. No other fleet-availability row is needed.

Idle days. A train is idle on day k when its assignment row is all zeros: $x_{ijk} = 0$ for every $j \in \{0, 1, \ldots, M\}$. An idle train flies no mission and is not in the depot. Idle days and depot days add no increment, because every day aggregate $\delta\mu_{i\ell k}$, $\delta v_{i\ell k}$, $\delta R_{i\ell k}$, $\delta K_{i\ell k}$ sums over missions $j \geq 1$ only. A depot day must be chosen explicitly ($x_{i0k} = 1$), and repairs and replacements need one (Section 5.2). Because Eq. (4) is an inequality, the choice between a depot day and an idle day is free, and the term $C_M \sum x_{i0k}$ of the objective prices it.

Depot-Day Compatibility

A component may be repaired or replaced only on a depot day of its train, and at most one of the two on the same day:

$$x^m_{i\ell k} + x^r_{i\ell k} \leq x_{i0k} \qquad \forall\, i, \ell, k. \tag{6}$$

With `allow_replacement = false` the row reads $x^m_{i\ell k} \leq x_{i0k}$. Several components of one train may receive interventions on the same depot day. A depot day without an intervention on a component leaves that component unchanged.

Remark (one row instead of three). Eq. (6) has the same binary solutions as the three rows $x^m \leq x_{i0k}$, $x^r \leq x_{i0k}$, $x^m + x^r \leq 1$. Its linear relaxation is tighter: the point $x_{i0k} = x^m = x^r = 0.5$ satisfies the three rows but not Eq. (6). With $0 \leq x^m, x^r$ and $x_{i0k} \leq 1$, Eq. (6) describes the convex hull of the four binary solutions $(x_{i0k}, x^m, x^r) \in \{(0,0,0), (1,0,0), (1,1,0), (1,0,1)\}$.

Repair Laws

An imperfect repair (in short, a repair) of a component on a depot day removes part of its degradation. The repair efficiency $\rho_{i\ell} \in (0, 1]$ is the fraction removed. Let D^- be the degradation level before the repair (at the end of day $k-1$; a depot day adds no increment) and D^+ the level after it (day k). The input `repair_model` selects one of two laws per cell; the default is `ardinf`.

Definition 5.1 (Repair laws)**.** Let $\rho = \rho_{i\ell}$. The two repair laws are:

- ARD_∞ (input value `ardinf`) removes the fraction ρ of the whole level:

$$D^+ = (1 - \rho)\, D^-. \tag{7}$$

- ARD_1 (input value `ard1`) removes the fraction ρ of the degradation accumulated since the last intervention (repair or replacement):

$$D^+ = D^- - \rho \left(D^- - D^{\mathrm{last}}\right), \tag{8}$$

  where D^{last} is the level just after the last intervention. Before the first intervention of the horizon, D^{last} is the level just after the last intervention before the horizon.

Names and sources. ARD_1, "arithmetic reduction of deterioration of order 1", is the law of [2, §3.1, eq. (4)]. That source measures the removed degradation from the level just after the last maintenance, as Eq. (8) does. The name ARD_∞ is an analogy, not a citation. Doyen and Gaudoin define the classes ARA_∞ [1, eq. (20), p. 11] and ARA_1 [1, eq. (22), p. 12], which reduce a virtual age, not a degradation level. ARA_∞ reduces the whole virtual age; ARA_1 reduces only the age accumulated since the previous repair. The subscript is the memory of the class: the largest number of previous failure times that enter the failure intensity [1, p. 8]. ARD_∞ applies the rule of ARA_∞ to the whole degradation level. No law of this specification acts on a virtual age.

Repair maps of the tracked quantities. A model tracks moments or other summaries of D, not D itself. For a tracked quantity s with anchor g, a repair on day k applies the repair map

$$\psi(s_{k-1}, g_{k-1}) = \begin{cases} a\, s_{k-1} & \mathrm{ARD}_\infty, \\ a\, s_{k-1} + (1-a)\, g_{k-1} & \mathrm{ARD}_1, \end{cases} \tag{9}$$

with a repair factor $a \in [0, 1]$. For the mean, $a = 1 - \rho_{i\ell}$, because expectation is linear: Eq. (8) gives $\mathbb{E}[D^+] = (1-\rho)\,\mathbb{E}[D^-] + \rho\,\mathbb{E}[D^{\mathrm{last}}]$. For a variance and for R the factor is $(1-\rho_{i\ell})^2$, and for K it is $1 - \rho_{i\ell}$; the model sections derive each factor. The Wiener variance has the factor $a = 1$ under both laws: a repair acts on the Wiener mean only and leaves the variance unchanged.

Anchors. Under ARD_1 each tracked quantity has an anchor g: the value of the quantity just after the last intervention. The Wiener variance is the exception: with $a = 1$ its map does not use an anchor, and none is built. The anchor recursion is

$$g_0 = \text{starting anchor}, \qquad g_k = \begin{cases} s_k & \text{on a repair or replacement day}, \\ g_{k-1} & \text{otherwise}, \end{cases} \tag{10}$$

built by the rows Eq. (14) or Eq. (17). The starting anchors $g^{\mu}_{i\ell 0}$, $g^{v}_{i\ell 0}$, $g^{R}_{i\ell 0}$, $g^{K}_{i\ell 0}$ come from the optional inputs `anchor_mu_0`, `anchor_v_0`, `anchor_R_0`, `anchor_K_0`. The default of each is the initial value of its quantity. Each must satisfy $0 \le g_0 \le s_0$ (Section **??**). Example: $\mu_{k-1} = 0.6$, $g_{k-1} = 0.2$, $\rho = 0.5$. A repair gives $\mu_k = 0.6 - 0.5\,(0.6 - 0.2) = 0.4$ and $g_k = 0.4$. On the next day the gap $\mu - g$ is 0, and it grows again with each mission.

Remark (default of the starting anchor). Damage only grows between interventions. So the true level after the last intervention before the horizon lies between the level of a new part and the initial level. A larger anchor gives a weaker repair and a higher state, because ψ is non-decreasing in g. The default $g_0 = s_0$ therefore never overstates what a repair achieves; the input gives the exact value when the maintenance history is known. Example: $\mu_0 = 0.5$, $\rho = 0.5$, and a repair on day 1 before any mission. With the starting anchor 0 the mean drops to 0.25; with the default μ_0 it stays at 0.5, and the repair removes only damage accumulated within the horizon.

Laws per model. Every model accepts both laws. Table 3 gives the reason for each model.

Table 3: Repair laws accepted by each model.

Model	ARD_∞	ARD_1	Reason or note
Gaussian	yes	yes	both laws keep the state Gaussian, with exact moments
Inverse Gaussian	yes	yes	the cap holds for any repair weights and any dependence between the terms (Section **??**)
Wiener	yes	yes	on the mean only; the variance changes only by replacement
Gamma	yes	yes	Section **??**

Rainflow, every bound	yes	yes	Chernoff under ARD_1 uses a conservative recursion for K (Section **??**)

Encoding of the Dynamics

Each day, every tracked quantity of a cell follows one rule: a mission adds an increment, a repair applies the repair map of Section 5.3, and a replacement resets the quantity (Section 5.5). This subsection states these rules once, for every model and both repair laws. The model sections give, for each tracked quantity, its increment, its repair factor and its natural range.

One-sided rows. Every dynamics row is an inequality ($\geq$), not an equality. The solver may choose a value above the result of the rule, never a value below it. So a tracked quantity is an upper bound on the value that the rule gives. Example: with $\mu_{i\ell,k-1} = 0.3$ and a mission increment of 0.2, the rule gives 0.5; the rows accept $\mu_{i\ell k} = 0.5$ and also $\mu_{i\ell k} = 0.6$. Replacement rows and anchor updates are equalities.

Generic rows. Let s be one tracked quantity of a cell (a mean, a variance, R or K), with day increment δs_k, natural range M_s, replacement value s^{new}, anchor g (law ARD_1 only) and repair map ψ (Section 5.3). Write x^m_k, x^r_k for the binaries of the cell and $e_k = x^m_k + x^r_k$. The input `encoding` selects one of two forms per cell. Both forms are built for $k = 1, \ldots, T$; index $k = 0$ holds the initial value and the starting anchor.

Big-M form (`encoding = "bigm"`, the default):

$$\text{no intervention:} \qquad s_k \geq s_{k-1} + \delta s_k - M_s\,(x^m_k + x^r_k), \tag{11}$$

$$\text{repair:} \qquad s_k \geq \psi(s_{k-1}, g_{k-1}) - M_s\,(1 - x^m_k), \tag{12}$$

$$\text{replacement:} \qquad |s_k - s^{\mathrm{new}}| \leq M_s\,(1 - x^r_k), \tag{13}$$

$$\text{anchor } (\mathrm{ARD}_1): \qquad |g_k - s_k| \leq M_s\,(1 - e_k), \qquad |g_k - g_{k-1}| \leq M_s\,e_k. \tag{14}$$

Each row with an absolute value is built as two linear rows.

Indicator form (`encoding = "indicator"`). One extra binary $n_{i\ell k}$ per cell and day marks a day without intervention:

$$n_{i\ell k} + x^m_{i\ell k} + x^r_{i\ell k} = 1. \tag{15}$$

The rows are Gurobi indicator constraints:

$$n_k = 1 \;\Rightarrow\; s_k \geq s_{k-1} + \delta s_k, \qquad x^m_k = 1 \;\Rightarrow\; s_k \geq \psi(s_{k-1}, g_{k-1}), \qquad x^r_k = 1 \;\Rightarrow\; s_k = s^{\mathrm{new}}, \tag{16}$$

$$n_k = 1 \;\Rightarrow\; g_k = g_{k-1}, \qquad x^m_k = 1 \;\Rightarrow\; g_k = s_k, \qquad x^r_k = 1 \;\Rightarrow\; g_k = s_k. \tag{17}$$

The binary $n_{i\ell k}$ is built only at cells in this form.

Common rules for both forms.

- The anchor rows are built only under ARD_1, one anchor per tracked quantity, except the Wiener variance.
- The Wiener variance takes the repair map $\psi = s_{k-1}$ (factor $a = 1$) under both laws, and has no anchor. A repair day adds no increment, so in the big-M form its no-intervention and repair rows together equal the single row $s_k \geq s_{k-1} + \delta s_k - M_s\,x^r_k$ (Eq. (**??**)), which may replace them. The variance is never left free below on a repair day.
- With `allow_replacement = false` the binaries x^r are not built: read $x^r_k = 0$ in every row, and drop the replacement rows.
- At binary points the two forms have the same feasible set. Every statement below holds for both.

Natural ranges. The big-M constant M_s is the natural range of s: an upper bound of s at every point that satisfies the reliability row of the cell. It is $\tau_{i\ell}$ for a mean and for its anchor, $V_{\max}$ for a Gaussian or Wiener variance, V^{C} or V^{B} for a rainflow variance (Cantelli or Bernstein), $2\tau^2/\kappa$ for R and $s_{i\ell}\tau - \kappa$ for K (Section **??**). An anchor has the range of its quantity.

Remark (validity of the big-M constants). Every tracked quantity and every anchor lies in $[0, M_s]$. The initial values, the starting anchors and the replacement values lie in $[0, M_s]$ too, by the input checks of Section **??**. An intervention day is a depot day (Section 5.2), so $\delta s_k = 0$ on it. A relaxed row then cuts no point: Eq. (11) with $e_k = 1$ reads $s_k \geq s_{k-1} - M_s$, whose right-hand side is at most 0; Eq. (12) with $x_k^m = 0$ has a right-hand side $\psi - M_s \leq 0$, because ψ is a convex combination of values in $[0, M_s]$; Eq. (13) with $x_k^r = 0$ allows $s_k \in [s^{\mathrm{new}} - M_s,\, s^{\mathrm{new}} + M_s] \supseteq [0, M_s]$; the relaxed anchor rows allow every difference in $[-M_s, M_s]$.

Remark (choice of the default form). Gurobi accepts a binary as integral within its integrality tolerance `IntFeasTol` (default 10^{-5}). A big-M row can therefore leak by at most $10^{-5} M_s$. Because M_s is the quantity's own range, this is a relative error of 10^{-5}. Indicator constraints have no such leak, and Gurobi usually needs more time to solve them. Which form solves this model faster has not been measured.

Safety of the one-sided rows

Fix a schedule, that is, values of every binary, and one cell. The following objects are used here and in Section 5.9.

- The *full tracked state* ξ_k collects the values on day k of every tracked quantity of the cell and of every anchor. ξ_0 holds the initial values and the starting anchors.
- The *day map* φ_k gives, for each coordinate, the value that the rule of day k assigns: $s + \delta s_k$ and g on a day without intervention; $\psi(s, g)$ for both s and g on a repair day; s^{new} for both on a replacement day.
- The *reference trajectory* is $\zeta_0 = \xi_0$, $\zeta_k = \varphi_k(\zeta_{k-1})$: the rows with every inequality replaced by an equality.
- The *true statistics* θ_k are the true values on day k of the tracked quantities (the mean, the variance, R or the CGF of the damage $D_{i\ell k}$) and of their anchors (their values just after the last intervention).
- The *certificate set* $\mathcal{C}_{i\ell}$ is the set of values of the tracked quantities at which the reliability row of the cell holds in its exact form, side rows included. Anchors do not enter it.

Every model section states three facts, labelled (M1)–(M3) there:

(M1) Each repair factor a lies in $[0, 1]$ and each increment δs_k is non-negative.

(M2) On day 0 the true statistics are at most the initial values and starting anchors, $\theta_0 \leq \xi_0$. On every later day the true statistic is at most the day map applied to the true statistics of the day before: $\theta_k \leq \varphi_k(\theta_{k-1})$. Every map holds with equality except the map of the Chernoff quantity K, which is an upper bound (Section **??**).

(M3) If values $s \in \mathcal{C}_{i\ell}$ are at least the true statistics of day k, then $\Pr(D_{i\ell k} > \tau_{i\ell}) \leq \varepsilon_{i\ell}$.

The set $\mathcal{C}_{i\ell}$ is closed, and it is a down-set: if $s \in \mathcal{C}_{i\ell}$ and $0 \leq s' \leq s$ componentwise, then $s' \in \mathcal{C}_{i\ell}$. The reason is that every reliability row, written in chance-bound form, reads "a continuous function of the tracked quantities, non-decreasing in each of them, is at most a constant":

Model or bound	Row in chance-bound form	Section
Gaussian, Wiener	$\mu + \Phi^{-1}(1-\varepsilon)\sqrt{v} \leq \tau$	**??**, **??**
Inverse Gaussian; Gamma	$\mu \leq \bar{\mu}$; $\mu \leq A_1(\beta^*\tau, \varepsilon)/\beta^*$	**??**, **??**
Rainflow, Markov	$\mu \leq \varepsilon\,\tau$	**??**
Rainflow, Cantelli	$\mu + \kappa_C\sqrt{v} \leq \tau$	**??**

Rainflow, Hoeffding	$\mu + \sqrt{\kappa R/2} \le \tau$	??
Rainflow, Bernstein	$\mu + c + \sqrt{c^2 + 2\kappa v} \le \tau$	??
Rainflow, Chernoff	$K \le s_{i\ell}\,\tau + \ln \varepsilon$	??

Lemma 5.2 (Monotone day maps)**.** *Each day map φ_k is non-decreasing in each coordinate of the full tracked state, anchors included.*

Proof. Let $\xi \le \xi'$ componentwise. Each coordinate of φ_k has one of the forms $s + \delta s_k$, g, s, $a\,s$, $a\,s + (1-a)\,g$ or s^{new}. By (M1), $a \ge 0$ and $1 - a \ge 0$. Therefore each coordinate of $\varphi_k(\xi)$ is at most the same coordinate of $\varphi_k(\xi')$. □

Proposition 5.3 (Domination)**.** *Every feasible solution satisfies $\xi_k \ge \varphi_k(\xi_{k-1})$ and $\theta_k \le \zeta_k \le \xi_k$ for $k = 1, \dots, T$.*

Proof. At binary points the rows of day k read: $s_k \ge \varphi_k(\xi_{k-1})_s$ for every tracked quantity s, with equality on a replacement day; $g_k = s_k \ge \varphi_k(\xi_{k-1})_g$ on an intervention day; $g_k = g_{k-1} = \varphi_k(\xi_{k-1})_g$ on any other day. So $\xi_k \ge \varphi_k(\xi_{k-1})$. Let $\theta_{k-1} \le \zeta_{k-1} \le \xi_{k-1}$ (true for $k = 1$ by (M2), since $\zeta_0 = \xi_0$). Then, by (M2) and Lemma 5.2,

$$\theta_k \le \varphi_k(\theta_{k-1}) \le \varphi_k(\zeta_{k-1}) = \zeta_k \le \varphi_k(\xi_{k-1}) \le \xi_k.$$

Therefore $\theta_k \le \zeta_k \le \xi_k$ for every k by induction. □

Consequence on the horizon. On every day $k = 1, \dots, T$ the tracked values satisfy the reliability row in the form chosen for the cell. Every form lies inside $\mathcal{C}_{i\ell}$ (Section 5.6), so $\xi_k \in \mathcal{C}_{i\ell}$. By Proposition 5.3, $\theta_k \le \xi_k$, so (M3) gives $\Pr(D_{i\ell k} > \tau_{i\ell}) \le \varepsilon_{i\ell}$. Theorem 5.7 (Section 5.9) extends this to every day of every later repetition of the operating phase.

What the one-sided rows accept. Call a schedule *safe* for a cell when the reference trajectory of the endless schedule (the transitory phase once, then the operating phase repeated without end, Section 5.9) lies in $\mathcal{C}_{i\ell}$ and in the natural ranges on every day. The one-sided rows relate to this property as follows.

- *They never accept an unsafe schedule* (Theorem 5.7).
- *They accept every schedule that equality rows accept*, and more. Equality rows force $\xi_k = \zeta_k$; Proposition 5.4 (a) shows that such a schedule satisfies the one-sided rows too. The first example below shows a safe schedule that only the one-sided rows accept.
- *They may reject a safe schedule*, as equality rows may, but only when tracked quantities move in opposite directions over the first repetition (Proposition 5.4 (b) and the second example below).

Proposition 5.4 (Acceptance)**.** *Fix a schedule and a cell, use the exact reliability form, and let the schedule be safe for the cell. Let $\bar{\zeta} = \zeta_{H_1}$ and let Φ be the composition of the day maps of the operating phase (Section 5.9).*

(a) If $\Phi(\bar{\zeta}) \le \bar{\zeta}$, the rows of the cell, the loop row included, accept the schedule with $\xi_k = \zeta_k$ for every k.

(b) If $\Phi(\bar{\zeta}) \ge \bar{\zeta}$, the cell has no anchor (law ARD_∞), and the cell is not replaced on day H_1, the rows of the cell accept the schedule.

Proof. (a) The values $\xi_k = \zeta_k$ satisfy every row of day k with equality at binary points. They lie in $\mathcal{C}_{i\ell}$ because the schedule is safe. The loop row reads $\zeta_T = \Phi(\bar{\zeta}) \le \bar{\zeta} = \zeta_{H_1}$.

(b) Let $y_n = \Phi^n(\bar{\zeta})$, the reference state at the end of the n-th repetition. By Lemma 5.2 and $\Phi(\bar{\zeta}) \ge \bar{\zeta}$, $y_{n+1} = \Phi(y_n) \ge \Phi(y_{n-1}) = y_n$. The sequence is non-decreasing and lies in the natural ranges, so it converges to some y^* with $y^* \ge \bar{\zeta}$. Each day map is affine or constant, so Φ

and its prefixes Φ_h are continuous: $\Phi(y^*) = \lim_n y_{n+1} = y^*$ and $\Phi_h(y^*) = \lim_n \Phi_h(y_n)$. Each $\Phi_h(y_n)$ is a reference value of a later day, so it lies in the closed set $\mathcal{C}_{i\ell}$; hence $\Phi_h(y^*) \in \mathcal{C}_{i\ell}$ for $h = 1, \dots, H_2$, and $y^* = \Phi_{H_2}(y^*) \in \mathcal{C}_{i\ell}$. Take $\xi_k = \zeta_k$ for $k < H_1$, $\xi_{H_1} = y^*$ and $\xi_{H_1+h} = \Phi_h(y^*)$. Day H_1 is not a replacement day and the cell has no anchor, so $\xi_{H_1} = y^* \geq \varphi_{H_1}(\xi_{H_1-1})$ satisfies the rows of day H_1. The later days satisfy their rows with equality, and the loop row reads $\xi_T = \Phi(y^*) = y^* = \xi_{H_1}$. □

A scalar state is always comparable with its image. So a cell with one tracked quantity under ARD_∞, not replaced on day H_1, accepts every safe schedule. Under ARD_1 the anchor is a second tracked quantity, and its value on day H_1 is fixed by the anchor rows; part (b) does not apply, and acceptance of every safe schedule is not proven.

Example (one quantity, accepted only by the one-sided rows). One tracked mean, law ARD_∞, $\rho = 0.5$. The operating phase is a mission ($\delta\mu = 0.2$) followed by a repair, so $\Phi(y) = 0.5\,(y+0.2) = 0.5\,y + 0.1$, and $\Phi(y) \leq y$ if and only if $y \geq 0.2$. Let $\bar{\zeta} = 0.1$. Then $\Phi(0.1) = 0.15 > 0.1$, so equality rows reject the schedule. The one-sided rows accept it with $\xi_{H_1} = 0.2$: the tracked cycle is $0.2, 0.4, 0.2$, and the reliability row is checked at 0.4. The reference trajectory is $0.1, 0.3, 0.15, 0.35, 0.175, \dots$; its start-of-cycle values rise to 0.2 and its peaks rise to 0.4. So the schedule is safe whenever $0.4 \in \mathcal{C}_{i\ell}$. The price is in the reported cost: each repair is charged $\rho \cdot 0.4 = 0.2$, while the reference removal rises from 0.15 toward 0.2.

Example (two quantities that move in opposite directions, rejected). Cantelli bound, $\varepsilon = 0.1$ ($\kappa_C^2 = 9$), $\tau = 1$, $\rho = 0.5$, law ARD_∞. The operating phase is a mission, which adds 0.2 to the mean and 0.009 to the variance, followed by a repair. So $\Phi(\mu, v) = (0.5\,\mu + 0.1,\ 0.25\,v + 0.00225)$, with fixed point $(0.2, 0.003)$. Let $\bar{\zeta} = (0.5, 0)$.

- Reference: over the repetitions the mean falls to 0.2 and the variance rises to 0.003. The first peak is $(0.7, 0.009)$, with $9 \cdot 0.009 = 0.081 \leq 0.3^2 = 0.09$. Every later peak has a mean of at most 0.55 and a variance below 0.012, with $9 \cdot 0.012 = 0.108 \leq 0.45^2 = 0.2025$. The schedule is safe.
- One-sided rows: the loop row needs $\xi_{H_1} = y \geq (0.5, 0)$ with $\Phi(y) \leq y$, so $y_v \geq 0.003$. The tracked peak is then at least $(0.7, 0.012)$, where $9 \cdot 0.012 = 0.108 > 0.09$. Rejected.
- Equality rows: $\Phi(\bar{\zeta}) = (0.35, 0.00225)$ has a larger variance than $\bar{\zeta}$. Rejected as well.

Meaning of the reported values. The output reports the tracked values ξ_k, not the reference values. By Proposition 5.3 each reported state and anchor is an upper bound on its reference value, and so on its true statistic. The costs computed from them are upper bounds too: the repair cost (Proposition 5.6) and the damage penalty (Section 5.8).

Remark (slack in the reported values). The minimisation removes slack only where slack costs something: a larger mean raises the damage penalty $C_D\,u$ where u is binding, and raises the repair cost $C_R\,z$ of every later repair. A larger variance, R or K tightens the reliability row and so restricts the schedule. Elsewhere the solver may return any value above the reference value. With $C_D = 0$ the damage penalty is absent (Section 5.8), and the reported states may carry more slack. Slack never affects safety (Theorem 5.7).

Replacement

A replacement installs a new part on a depot day. The input `allow_replacement` (default `true`) makes replacements available. When it is `false`, the binaries $x^r_{i\ell k}$, the replacement rows and the replacement cost are not built.

State after a replacement. On a replacement day every tracked quantity of the cell takes its replacement value, and every anchor takes the same value (Eq. (10)). The values come from the input, for every model:

- mean $\mu_{i\ell}^{\text{new}}$, input `replacement_mu`, default 0;
- variance $v_{i\ell}^{\text{new}}$, input `replacement_v`, default 0, where the model tracks a variance;
- $R_{i\ell}^{\text{new}}$ and $K_{i\ell}^{\text{new}}$ for the rainflow Hoeffding and Chernoff bounds, with their defaults and their derivation in Section **??**.

A replacement value describes the damage of the new part: a known damage when its variance is 0, a random one otherwise. The rows are the equalities Eq. (13) or the indicator rows Eq. (16). The input checks of Section **??** require each replacement value to lie in the natural range of its quantity, and the replacement state to satisfy the reliability row of the cell; otherwise no replacement of that cell would be feasible.

Shortcut for a new part. The input `replacement_as_new` (default `false`) states that a replaced part is new, with no damage: every replacement value of the cell is 0. A non-zero replacement input together with `replacement_as_new = true` is an input error (Section **??**).

Cost. A replacement costs $C_{\text{rep},i\ell}$ and nothing else: the repair cost $z_{i\ell k}$ is 0 on a replacement day (Section 5.7). A cell may be replaced on any depot day of its train, any number of times. The loop row can require at least one replacement in the operating phase: for every ARD_1 cell with $\rho < 1$ that accumulates damage (Proposition 5.8), and for the Wiener variance (Section **??**).

Linear Forms of the Reliability Rows

Five reliability rows are quadratic: the Gaussian and Wiener rows, and the rainflow Cantelli, Hoeffding and Bernstein rows. Each has the form

$$\chi_{i\ell k} \;\le\; f(\mu_{i\ell k}), \qquad f(\mu) = c_2\,(\tau-\mu)^2 + c_1\,(\tau-\mu), \qquad \mu_{i\ell k} \le \bar{\mu}_{\text{side}}, \tag{18}$$

with the constants of Table 5. The third relation is the side row; it selects the branch of the parabola on which the row is a valid certificate. Since $c_2 > 0$, f is convex in μ, with $f'(\mu) = -2c_2(\tau-\mu) - c_1$.

Table 5: Constants of the quadratic reliability rows, Eq. (18). The model sections derive each row.

Row	χ	c_2	c_1	$\bar{\mu}_{\text{side}}$
Gaussian, Wiener	v	$1/(\Phi^{-1}(1-\varepsilon))^2$	0	τ
Rainflow, Cantelli	v	$1/\kappa_C^2 = \varepsilon/(1-\varepsilon)$	0	τ
Rainflow, Hoeffding	R	$2/\kappa$	0	τ
Rainflow, Bernstein	v	$1/(2\kappa)$	$-b_{i\ell}/3$	$\tau - c_{i\ell}$

The input `reliability_impl` selects one of three forms per cell. Every form keeps the side row.

- `exact`: Eq. (18) as written. The feasible set $\{(\mu,\chi) : \chi \le f(\mu)\}$ lies below a convex function, so it is not convex: the row is a nonconvex quadratic constraint. Gurobi's parameter `NonConvex` is set to 2 whenever at least one cell has a quadratic row in this form.
- `tangent`: one linear row per day, the tangent of f at the point $\mu_r = \texttt{tangent_ref}_{i\ell} \cdot \tau_{i\ell}$:

$$\chi_{i\ell k} \;\le\; f(\mu_r) + f'(\mu_r)\,(\mu_{i\ell k} - \mu_r). \tag{19}$$

  The input `tangent_ref` is per cell, in $[0,1)$, default 0.5. At $\texttt{tangent_ref} = 0$ the Gaussian row becomes $(\Phi^{-1}(1-\varepsilon))^2\, v + 2\tau\mu \le \tau^2$. The value 1 is excluded: for $c_1 = 0$ the tangent at τ is $\chi \le 0$. At a Bernstein cell the tangent point must also satisfy $\mu_r < \tau - c_{i\ell}$ (Section **??**); otherwise no point with $v \ge 0$ satisfies the row (Section **??**).

- `pwl`: a piecewise tangent with $P =$ `pwl_points` segments (default 8). The breakpoints split $[0, \tau]$ into equal segments $[(p-1)\tau/P,\ p\tau/P]$ with midpoints $\hat{\mu}_p = (p - \frac{1}{2})\tau/P$, $p = 1, \dots, P$. One binary $w_{i\ell kp}$ per cell, day and segment selects the segment that contains the mean:

$$\sum_{p=1}^{P} w_{i\ell kp} = 1, \tag{20}$$

$$w_{i\ell kp} = 1 \ \Rightarrow\ (p-1)\,\tau/P \le \mu_{i\ell k} \le p\,\tau/P, \tag{21}$$

$$w_{i\ell kp} = 1 \ \Rightarrow\ \chi_{i\ell k} \le f(\hat{\mu}_p) + f'(\hat{\mu}_p)\,(\mu_{i\ell k} - \hat{\mu}_p). \tag{22}$$

  The implications are Gurobi indicator constraints, whatever the value of `encoding`. With $P = 1$ the form is the single tangent at $\tau/2$, with no binaries.

At a cell whose reliability row is linear (Markov, Chernoff, inverse Gaussian, Gamma) the row is built in its exact form, and the value of `reliability_impl` is ignored. An unknown value of `reliability_impl` is an input error at every cell (Section **??**).

Proposition 5.5 (Safety of the linear forms)**.** *For every μ_r and every μ, $f(\mu_r) + f'(\mu_r)\,(\mu - \mu_r) = f(\mu) - c_2\,(\mu - \mu_r)^2 \le f(\mu)$. Therefore every point that satisfies the* ***`tangent`*** *or the* ***`pwl`*** *form satisfies the* ***`exact`*** *form, and both linear forms lie inside the certificate set $\mathcal{C}_{i\ell}$.*

Proof. f is a polynomial of degree 2 in μ with leading coefficient c_2, so its Taylor expansion at μ_r is exact: $f(\mu) = f(\mu_r) + f'(\mu_r)(\mu - \mu_r) + c_2(\mu - \mu_r)^2$. Since $c_2 > 0$, the tangent lies below f. Each linear form caps χ by one such tangent and keeps the side row. Therefore it implies Eq. (18). □

Remark (accuracy). The `tangent` form loses $c_2(\mu - \mu_r)^2$ in χ: it is exact at $\mu = \mu_r$ and loosest far from it. A tangent point near the operating mean of the cell is the best choice. In the `pwl` form the loss is $c_2(\mu - \hat{\mu}_p)^2 \le c_2\,\tau^2/(4P^2)$. The midpoint of the segment that contains μ is the nearest tangent point to μ. So the `pwl` form accepts exactly the points with $\chi \le \max_p [f(\hat{\mu}_p) + f'(\hat{\mu}_p)(\mu - \hat{\mu}_p)]$, the largest set that the P tangents allow. The price is P binaries per cell and day.

Example (Cantelli, $\varepsilon = 0.1$, $\kappa_C^2 = 9$, $\tau = 1$, $v = 0.004$). The largest admissible mean is $1 - \sqrt{9 \cdot 0.004} \approx 0.810$ with `exact`; 0.482 with `tangent` at 0 ($9v + 2\mu \le 1$); 0.714 with `tangent` at $0.5\,\tau$ ($9v \le 0.75 - \mu$); and 0.810 with `pwl` and $P = 8$. The tangent at 0 never admits a mean above $\tau/2$, even with zero variance.

Repair Cost Variable

The variable $z_{i\ell k} \ge 0$ is the degradation removed by a repair of cell (i, ℓ) on day k, in mean units, for every model and both laws. The objective charges $C_{R,i\ell}\,z_{i\ell k}$ (Section 4). The variable and its rows are built only at cells with $C_{R,i\ell} > 0$, and only on the days whose cost enters the objective: every day $k = 1, \dots, T$, except the days of the transitory phase in the mode `operating_average`. The rows are

$$z_{i\ell k} \ \ge\ \rho_{i\ell}\left(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1}\right) - \rho_{i\ell}\,\tau_{i\ell}\left(1 - x^{m}_{i\ell k}\right), \qquad z_{i\ell k} \ge 0, \tag{23}$$

with $g^{\mu} \equiv 0$ under ARD_{∞}. On a repair day the first row reads $z_{i\ell k} \ge \rho\,(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1})$, the removed amount of Eq. (9). On any other day its right-hand side is at most 0, because $0 \le \mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1} \le \mu_{i\ell,k-1} \le \tau$.

Replacement days. A replacement day has $x^{m}_{i\ell k} = 0$, so $z_{i\ell k} = 0$ at the optimum. A replacement costs $C_{\mathrm{rep},i\ell}$ only. The replaced part is discarded, so its damage does not change the price of the new part.

Remark (no upper rows). The variable $z_{i\ell k}$ appears in no row other than Eq. (23), and in the objective with a positive coefficient. The minimisation therefore drives it to the largest of its

lower bounds: $z_{i\ell k} = \rho\,(\mu_{i\ell,k-1} - g^{\mu}_{i\ell,k-1})$ on a repair day and $z_{i\ell k} = 0$ on any other day. The upper rows of a McCormick envelope, $z \le \rho\,(\mu - g^{\mu})$ and $z \le \rho\,\tau\,x^{m}$, would only remove points that are never optimal, so they are not built. A solution that is not optimal (for example, the best one found at the time limit) may carry a larger z; its reported cost is then larger, never smaller.

Proposition 5.6 (What z measures)**.** *In every feasible solution, on every repair day k of a cell, under both laws,*

$$z_{i\ell k} \;\ge\; \rho\,\big(\mu_{k-1} - g^{\mu}_{k-1}\big) \;\ge\; \rho\,\big(\zeta^{\mu}_{k-1} - \zeta^{g}_{k-1}\big),$$

where ζ^{μ} and ζ^{g} are the mean and its anchor on the reference trajectory (Section 5.4). When the starting anchor equals the true mean level after the last intervention before the horizon, the right-hand side is the expected degradation that the repair removes.

Proof. The first inequality is Eq. (23) with $x^{m}_{i\ell k} = 1$. Under ARD_{∞}, $g^{\mu} = \zeta^{g} = 0$, and $\mu_{k-1} \ge \zeta^{\mu}_{k-1}$ by Proposition 5.3. Under ARD_{1}, let $t < k$ be the last intervention day before k, or $t = 0$ if there is none. Then $g^{\mu}_{k-1} = g^{\mu}_{t}$ and $\zeta^{g}_{k-1} = \zeta^{g}_{t}$.

- If $t = 0$: $g^{\mu}_{0} = \zeta^{g}_{0}$ (the starting anchor), so $g^{\mu}_{k-1} - \zeta^{g}_{k-1} = 0 \le \mu_{k-1} - \zeta^{\mu}_{k-1}$.
- If $t \ge 1$: $g^{\mu}_{t} = \mu_{t}$ and $\zeta^{g}_{t} = \zeta^{\mu}_{t}$ (Eq. (10)). On the days $j = t+1, \dots, k-1$ there is no intervention: $\mu_{j} \ge \mu_{j-1} + \delta\mu_{j}$ and $\zeta^{\mu}_{j} = \zeta^{\mu}_{j-1} + \delta\mu_{j}$. So $\mu_{j} - \zeta^{\mu}_{j}$ does not decrease, and $\mu_{k-1} - \zeta^{\mu}_{k-1} \ge \mu_{t} - \zeta^{\mu}_{t} = g^{\mu}_{k-1} - \zeta^{g}_{k-1}$.

Rearranging gives $\mu_{k-1} - g^{\mu}_{k-1} \ge \zeta^{\mu}_{k-1} - \zeta^{g}_{k-1}$. The mean recursion is linear (Section 5.3); with an exact starting anchor, ζ^{μ} and ζ^{g} are the true means, so $\rho\,(\zeta^{\mu}_{k-1} - \zeta^{g}_{k-1}) = \mathbb{E}[D^{-} - D^{+}]$. □

A starting anchor above the true one, which the default may be (Section 5.3), models a weaker repair. For the first repair before any intervention of the horizon, the charged removal then follows the starting anchor and may be below the true removal. From the first intervention on, the reference gap $\zeta^{\mu} - \zeta^{g}$ equals the true gap, the sum of the mean increments since that intervention, so the bound holds again.

Under both laws, then, the reported repair cost $\sum C_{R}\,z$ is an upper bound on the expected repair cost of the schedule, with the exception above for a starting anchor above the true one. Under ARD_{1} the tracked mean and the tracked anchor are both upper bounds, and their difference is still an upper bound on the reference difference: the anchor carries the slack that the mean had at the last intervention, and the slack of the mean does not decrease until the next one. The bound is exact when the tracked mean has no slack.

Damage Regularisation

The scalar $u \ge 0$ measures the largest damage of one train on one day, over the whole horizon. The objective charges $C_{D}\,u$ (Section 4). The variable u and its rows are built only when $C_{D} > 0$. With $C_{D} = 0$ they are not built, and the input `penalty_type` is ignored. The input check requires $C_{D} \ge 0$ (Section **??**): a negative C_{D} would reward damage.

The input `penalty_type` selects one of two forms. Both have one row per train and day.

- `inf_norm` (default), a linear row:

$$u \;\ge\; \sum_{\ell=1}^{L} \mu_{i\ell k} \qquad \forall\, i, k. \tag{24}$$

  At the optimum, $u = \max_{i,k} \sum_{\ell} \mu_{i\ell k}$.

- `quadratic`, a convex quadratic row:

$$u \;\geq\; \sum_{\ell=1}^{L} \mu_{i\ell k}^2 \qquad \forall\, i, k. \tag{25}$$

  The row is a rotated second-order cone, $u \cdot 1 \geq \|\mu_{i\cdot k}\|_2^2$, which Gurobi solves without the parameter `NonConvex`. It requires `reliability_impl = "exact"` at every cell (Section **??**): the linear forms are chosen for a model without quadratic rows, and this row is quadratic.

Role. The term prices the worst train-level damage, so the minimisation spreads damage across the fleet. It also removes slack from the tracked means where u is binding (Section 5.4). Since $\mu_{i\ell k} \geq \zeta^{\mu}_{i\ell k} \geq 0$ (Proposition 5.3), both forms of u are upper bounds on the same quantity computed from the reference values.

Loop Constraint

The schedule of the operating phase is meant to repeat without end. One hard row per tracked quantity and per anchor makes every repetition safe:

$$s_{i\ell T} \;\leq\; s_{i\ell H_1} \qquad \text{for every cell } (i,\ell) \text{ and every coordinate } s \text{ of its full tracked state.} \tag{26}$$

The rows cover every tracked quantity of the cell (the mean, and the variance, R or K where the model tracks them) and, under ARD_1, every anchor. There is no soft loop penalty in the objective.

Endless schedule. Repeat the operating phase without end: day $T + nH_2 + h$, for $n \geq 0$ and $1 \leq h \leq H_2$, carries the binaries of day $H_1 + h$. It also carries the increments of day $H_1 + h$, because the operating profile is read cyclically with period H_2 (Section **??**). So every repetition applies the same day maps $\varphi_{H_1+1}, \ldots, \varphi_T$. Write $\Phi_h = \varphi_{H_1+h} \circ \cdots \circ \varphi_{H_1+1}$ for the first h days of the operating phase and $\Phi = \Phi_{H_2}$ for the whole phase.

Theorem 5.7 (Perpetual safety)**.** *Let a solution satisfy every row of the problem. Then, for every cell and every day $k \geq 1$ of the endless schedule, the true statistics θ_k and the reference values ζ_k are at most a tracked value of the horizon that lies in $\mathcal{C}_{i\ell}$. Therefore $\zeta_k \in \mathcal{C}_{i\ell}$ and $\Pr(D_{i\ell k} > \tau_{i\ell}) \leq \varepsilon_{i\ell}$ on every day $k \geq 1$.*

Proof. Fix a cell. Let ξ_k be its tracked full state, and θ_k and ζ_k the true statistics and the reference trajectory of the endless schedule. By (M2) and Lemma 5.2, the induction of Proposition 5.3 gives $\theta_k \leq \zeta_k$ on every day $k \geq 1$.

- Days $k \leq T$: $\zeta_k \leq \xi_k$ (Proposition 5.3) and $\xi_k \in \mathcal{C}_{i\ell}$ (consequence on the horizon, Section 5.4).
- By Proposition 5.3 and Lemma 5.2, $\xi_{H_1+h} \geq \varphi_{H_1+h}(\xi_{H_1+h-1}) \geq \cdots \geq \Phi_h(\xi_{H_1})$ for $h = 1, \ldots, H_2$. With Eq. (26), $\Phi(\xi_{H_1}) \leq \xi_T \leq \xi_{H_1}$.
- Let $\Phi^n(\xi_{H_1}) \leq \xi_{H_1}$ (true for $n = 0$). Then $\Phi^{n+1}(\xi_{H_1}) = \Phi(\Phi^n(\xi_{H_1})) \leq \Phi(\xi_{H_1}) \leq \xi_{H_1}$ by Lemma 5.2. So $\Phi^n(\xi_{H_1}) \leq \xi_{H_1}$ for every $n \geq 0$.
- Day $k = H_1 + nH_2 + h$ with $n \geq 0$, $1 \leq h \leq H_2$:

$$\zeta_k = \Phi_h\big(\Phi^n(\zeta_{H_1})\big) \leq \Phi_h\big(\Phi^n(\xi_{H_1})\big) \leq \Phi_h(\xi_{H_1}) \leq \xi_{H_1+h} \in \mathcal{C}_{i\ell}.$$

Therefore, on every day $k \geq 1$, $\theta_k \leq \zeta_k \leq s$ for a tracked value $s \in \mathcal{C}_{i\ell}$. Since $\mathcal{C}_{i\ell}$ is a down-set, $\zeta_k \in \mathcal{C}_{i\ell}$, and (M3) gives $\Pr(D_{i\ell k} > \tau_{i\ell}) \leq \varepsilon_{i\ell}$. □

The theorem holds for every model and both repair laws, because it uses only the facts (M1)–(M3) and Lemma 5.2. It holds in both encodings and in every reliability form. It needs no relation between H_1 and H_2: the transitory phase may be shorter or longer than the operating phase,

because H_1 only marks the day at which the repeated block starts. Slack in the tracked values does not weaken it: the proof compares the reference trajectory with the tracked values, whatever slack they carry.

Remark (why the loop row covers the full state). The third step of the proof needs $\Phi(\xi_{H_1}) \le \xi_{H_1}$ in every coordinate, anchors included. Example: law ARD_1, $\rho = 0.5$, the first phase ends at $(\mu, g) = (0.4, 0.1)$, and the operating phase is a mission (+0.1), a repair, a mission (+0.05). The phase ends at $(0.35, 0.3)$, since the repair gives $0.5 \cdot 0.5 + 0.5 \cdot 0.1 = 0.3$. The row on the mean holds ($0.35 \le 0.4$); the row on the anchor fails ($0.3 > 0.1$). Each repetition raises the state after the repair by 0.075, so the mean grows without bound: the third repetition already peaks at 0.525, above the checked peak 0.5. A loop row on the mean alone would accept this unsafe schedule.

Remark (no loop row on a scalar bound). The loop row must compare the tracked quantities one by one. A single comparison of a scalar function of them, such as the failure-probability bound $U(\mu, v)$, does not chain: U takes the same value at many states, and its level sets are not invariant under the day maps. Example (Cantelli, $U = v/(v + (\tau - \mu)^2)$, $\tau = 10$, $\rho = 0.5$, law ARD_∞; the operating phase is one repair day followed by mission days that add 3 to the mean and 3.99 to the variance in total): from $(\mu, v) = (4, 9)$, $U = 0.2000$ at the end of the first phase and $U = 0.1997$ at the end of the operating phase, so a scalar row would accept. The next repetition gives $U = 0.2151 > 0.2$, and U tends to 0.2495. Eq. (26) rejects the schedule, since the mean rises from 4 to 5. The loop row states an order between tracked quantities; it is not a stochastic order between damage laws.

Proposition 5.8 (Replacement under ARD_1)**.** *Let a cell follow* ARD_1 *with* $\rho_{i\ell} < 1$*, and let its train fly, in the operating phase, at least one mission with a positive mean increment for the cell. Then every feasible solution has at least one replacement of the cell in the operating phase.*

Proof. Let the cell have no replacement in the operating phase. By the anchor rows, $g_k = g_{k-1}$ on a day without intervention and $g_k = \mu_k$ on a repair day. Every day satisfies $\mu_k \ge g_k$: on an intervention day with equality, and on other days because μ does not decrease and g stays constant (with $g_0 \le \mu_0$, Section 5.3). On a repair day k of the operating phase,

$$g_k = \mu_k \ge (1 - \rho)\, \mu_{k-1} + \rho\, g_{k-1} = g_{k-1} + (1 - \rho)\, (\mu_{k-1} - g_{k-1}).$$

So $g_T \ge g_{H_1} + (1 - \rho) \sum (\mu_{k-1} - g_{k-1})$, the sum over the repair days of the operating phase, each term non-negative. The anchor row $g_T \le g_{H_1}$ and $\rho < 1$ force every term to 0. Then every repair day gives $\mu_k \ge \mu_{k-1}$, every other day gives $\mu_k \ge \mu_{k-1} + \delta\mu_k$, and

$$\mu_T \ge \mu_{H_1} + \textstyle\sum_{k=H_1+1}^{T} \delta\mu_k > \mu_{H_1},$$

which violates the row $\mu_T \le \mu_{H_1}$. Therefore the cell is replaced at least once in the operating phase. □

The requirement is not an artefact of the rows. Without a replacement, each repair leaves the share $1 - \rho$ of the damage since the last intervention in place, so the reference mean grows without bound over the repetitions, as in the first remark above. With $\rho = 1$ a repair returns the state to the anchor, and no replacement is needed. The Wiener model needs a replacement in the operating phase for another reason, its variance (Section **??**).

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